Digitalizing Midwifery Care: AR and VR Solutions for Maternal Education

Project Proposal Report

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BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

Sri Lankan Institute of Information Technology

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Automated Fetal Health Risk Assessment System for Rural Midwifery Care using Lightweight Machine Learning

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Declaration, copyright statement and the statement of the supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Mr. Didula Thanaweera Arachchi

Signature of the supervisor:

Date: 8/29/2025

Abstract

This research proposes a mobile application designed to enhance midwifery practice during the antenatal (before birth) and intrapartum (during birth) phases. The application integrates maternal health monitoring, decision-support tools, localized educational resources, and data-driven dashboards for both students and practicing midwives.

Current maternal care apps (such as Pregnancy and BabyCenter) are primarily designed for mothers and do not address midwives' competency development. Similarly, while training apps such as the Safe Delivery App provide clinical guidelines, they lack cultural adaptation, real-time monitoring, and data analysis features relevant to Sri Lanka.

The proposed system will include tools to record and track vital maternal indicators such as blood pressure, blood sugar, and fetal heart rate, and will provide decision-support modules for delivery scenarios such as breech birth, postpartum hemorrhage, and neonatal resuscitation. A data analysis dashboard will support midwives and students by visualizing trends, providing alerts, and offering feedback on performance.

The novelty of this project is its Sri Lankan cultural adaptation with Sinhala, Tamil, and English support, ensuring accessibility and adoption across rural and urban communities. By combining mobile app development with data analytics, this project aims to strengthen maternal safety and midwifery competency training in Sri Lanka.

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List of Abbreviations

Abbreviation	Description
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CTG	Cardiotocography (The technical term for the fetal heart rate monitoring trace).
FHR	Fetal Heart Rate (The main parameter analyzed by the AI).
ML	Machine Learning (The core technology used for prediction).
AI	Artificial Intelligence.
XAI	Explainable Artificial Intelligence (The feature that gives reasons for the risk).
VOG	Visiting Obstetrician and Gynaecologist (The specialist doctor).
SMOTE	Synthetic Minority Over-sampling Technique (Used to balance the dataset).
UCI	University of California, Irvine (The source of the dataset).
ASTV	Abnormal Short Term Variability (A key feature used by the model to detect risk).
API	Application Programming Interface (Used to connect the Python model with the Mobile App).
RF	Random Forest (The specific algorithm used).
MOH	Ministry of Health (Sri Lanka).
FHB	Family Health Bureau (Sri Lanka).

Introduction

Background & Literature survey

This research proposes an automated Machine Learning (ML) based component designed to assist midwives in the early detection of fetal health risks during the antenatal period. In Sri Lanka's rural healthcare sector, while Cardiotocography (CTG) machines are available in peripheral units, the expert knowledge required to interpret complex fetal heart rate patterns is often limited to Obstetricians (VOGs), who are not always available. This delay in expert interpretation can lead to preventable adverse outcomes such as fetal hypoxia, cerebral palsy, or stillbirths.

Current automated analysis systems, such as "SisPorto," are primarily desktop-based and expensive, making them unsuitable for mobile midwifery units. Similarly, existing Deep Learning models often require high computational power and active internet connectivity, restricting their use in low-resource rural settings. Furthermore, most AI models function as "black boxes," providing predictions without clinical reasoning, which limits their adoption by medical staff.

The proposed system addresses these gaps by utilizing a lightweight **Random Forest Classifier** to analyze CTG parameters (e.g., baseline heart rate, accelerations, decelerations) and instantly classify fetal health into **Normal, Suspect, or Pathological** categories. A key novelty of this project is the integration of **Explainable AI (XAI)**, which provides diagnostic reasons for high-risk predictions (e.g., "Prolonged Decelerations Detected"), thereby building trust and aiding midwives in decision-making. Designed to function **offline on standard mobile devices**, this system aims to bridge the specialist gap in rural healthcare, ensuring timely interventions and strengthening maternal and fetal safety across Sri Lanka.

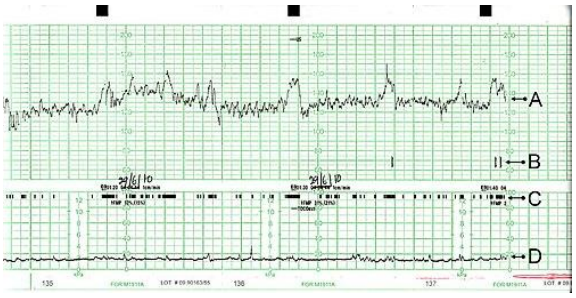


Figure 1 Cardiotocography Report

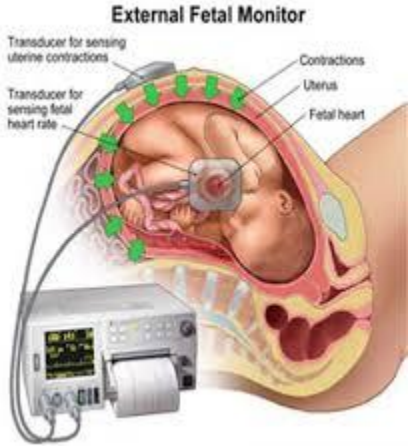


Figure 2 Cardiotocography

Research Gap

Several research studies on automated fetal health monitoring and Machine Learning (ML) applications have been conducted, but very few are directly applicable to **rural midwifery practice** in low-resource settings. Most existing automated systems are either designed for high-tech hospital environments (requiring expensive hardware) or rely on complex Deep Learning models that demand high computational power. While these systems achieve high accuracy, they lack the **portability, offline capability, and explainability** required for a mobile-based solution used by field midwives.

Research A – SisPorto 2.0 (Ayres-de-Campos et al., 2000) is a renowned computer-based system developed to analyze Cardiotocography (CTG) signals. It mimics expert interpretation with high clinical validity. However, it is a **desktop-based software** designed specifically for well-equipped hospitals with specialist doctors. It requires dedicated hardware and is not portable, making it unsuitable for rural midwives who conduct mobile clinics or home visits.

Research B – Deep Learning Approaches (e.g., Zhao et al., 2019) utilize Complex Neural Networks (CNNs) to classify fetal states with high precision. While scientifically advanced, these models suffer from high **computational load**, requiring powerful GPUs or cloud servers to function. Furthermore, they often operate as **"Black Boxes,"** providing a risk prediction without offering clinical reasons (explainability),

which reduces trust among medical staff who need to understand *why* a fetus is at risk.

The noticeable research gap is that while automated diagnostic tools exist, there are no comprehensive **mobile-based applications in Sri Lanka** that combine:

- **Lightweight Machine Learning:** Optimized algorithms (like Random Forest) capable of running **offline** on standard mobile smartphones without internet dependency.
- **Explainable AI (XAI):** Features that not only predict the risk but also provide **diagnostic reasoning** (e.g., identifying "Prolonged Decelerations") to support the midwife's decision-making.
- **Low-Cost Accessibility:** A solution designed specifically to bridge the specialist gap in **rural peripheral units** where VOGs are not immediately available.

Research Problem

Midwives in rural Sri Lankan hospitals face a critical **"Interpretation Gap"** when monitoring fetal wellbeing. While Cardiotocography (CTG) machines are available in many peripheral units, the advanced clinical expertise required to interpret complex fetal heart rate patterns (such as subtle decelerations or variability changes) is typically limited to Obstetricians (VOGs). In rural settings, a VOG is not always physically present, especially during night shifts or in emergencies.

Current practices rely heavily on the midwife's manual observation, which can be subjective and prone to human error. Existing automated analysis systems are restricted to desktop computers in major hospitals and are not portable. Consequently, there is a lack of digital tools that provide **instant, expert-level diagnostic support** on mobile devices. This delay in accurate diagnosis often leads to preventable adverse outcomes, such as fetal hypoxia, cerebral palsy, or stillbirths, due to the inability to detect distress signals in real-time.

Problem Statement

How can a lightweight, offline-capable Machine Learning model be engineered to automate the risk assessment of fetal health using CTG parameters, providing real-time, explainable diagnostic support for rural midwives in Sri Lanka?

Author / Existing System	**Methodology / Focus**	**Research Gap (Limitations)**
Ayres-de-Campos et al. (2000) *(The SisPorto 2.0 System)*	Developed a computer-based system to analyze CTG signals in hospitals, achieving high clinical accuracy acceptable to expert obstetricians.	**Desktop-based & Expensive:** Designed for high-tech hospitals with dedicated hardware. It is not portable and requires a specialist doctor to operate, making it unsuitable for rural mobile midwifery units.
Zhao et al. (2019) *(Deep*	Utilized Complex Neural Networks (CNN) and Deep Learning algorithms to	**High Computational Load:** These models require powerful GPUs or cloud servers. They are too "heavy" to run

Author / Existing System	Methodology / Focus	Research Gap (Limitations)
Learning on CTG)	classify fetal states with high precision.	offline on standard mobile phones used by field midwives and depend on active internet connectivity.
Ogasawara et al. (2021) *(Machine Learning Approaches)*	Applied standard ML models like SVM for fetal heart rate classification to automate diagnosis.	**"Black-Box" Issue:** These models provide a prediction (e.g., "Pathological") but fail to provide **Explainability** (clinical reasons). This lack of transparency reduces trust among medical staff who need to know *why* a fetus is at risk.
Maternity Foundation (2021) *(Safe Delivery App)*	A mobile app providing clinical guidelines and instructional videos for midwives to handle birth complications.	**Static Content Only:** While excellent for education, it lacks **Real-time Diagnostic Capabilities**. It cannot analyze patient data (CTG) or provide instant risk predictions for a specific mother.
Proposed System *(Automated Risk Assessment)*	**Lightweight Random Forest Classifier + Explainable AI (XAI)** running on a mobile architecture.	**Fills the Gap:** Provides a comprehensive solution that is **Mobile-First, Offline-Capable**, and **Explainable**. It brings expert-level diagnostics to low-resource rural settings without needing heavy hardware.

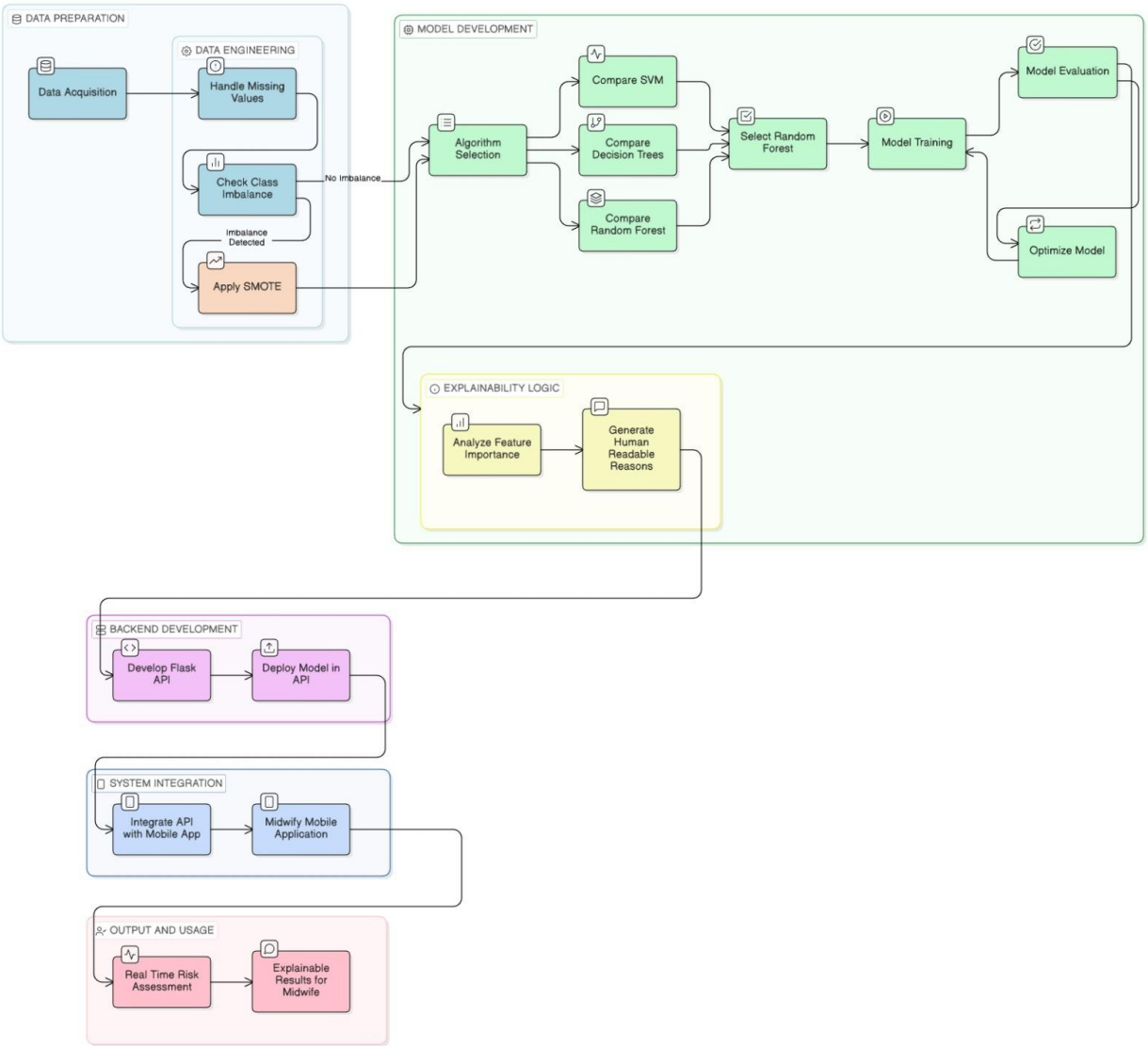
Objectives

Main Objectives

To engineer an automated, offline-capable Machine Learning component that classifies fetal health risks based on CTG parameters, providing real-time, explainable diagnostic support for rural midwives.

Specific Objectives

Figure 1 System Diagram



1. To analyze and preprocess clinical CTG data using **SMOTE** (Synthetic Minority Over-sampling Technique) to handle class imbalance and ensure high sensitivity in detecting pathological cases.
2. To train and optimize a lightweight **Random Forest Classifier** model capable of performing accurate risk assessment on mobile devices with limited computational resources.
3. To implement **Explainable AI (XAI)** logic to extract and display diagnostic reasons (e.g., "Prolonged Decelerations") for risk predictions, enhancing clinical trust.
4. To develop a secure **RESTful API** using Python Flask to integrate the predictive model with the main mobile application for real-time inference.
5. To validate the model's performance using standard evaluation metrics (Confusion Matrix, Recall, Precision) against the verified UCI clinical dataset.

Machine Learning Pipeline Development

- Data Engineering: Preprocessing using Python Pandas and Scikit-learn.
- Model Optimization: Hyperparameter tuning for lightweight performance.
- Explainability: Feature importance extraction for clinical reasoning.

Backend API Integration

- API Architecture: Python Flask microservice for model hosting.
- Communication: JSON-based data exchange between Mobile App and ML Model.
- Deployment: Offline-ready architecture for low-resource settings.

Methodology

The system development involves:

The development of the AI component follows the standard Data Science Lifecycle (DSLC):

1. **Data Acquisition & Analysis** – Retrieving the Fetal Health Dataset from the UCI Machine Learning Repository and analyzing clinical features.
2. **Data Preprocessing** – Cleaning data, handling missing values, and applying **SMOTE (Synthetic Minority Over-sampling Technique)** to balance the dataset for accurate risk detection.
3. **Model Engineering** – Training and optimizing the **Random Forest Classifier** to ensure high accuracy while remaining lightweight for mobile use.
4. **Backend API Development** – Creating a Python Flask API to host the trained model and enabling communication with the mobile app.
5. **Testing & Validation** – Validating the model using Confusion Matrices and conducting user acceptance testing (UAT) with potential users.

Table 2 Technologies Used

Technology	***Description***
Python (Pandas, Scikit-learn)	*Core programming language and libraries used for data preprocessing, machine learning model training, and evaluation.*
Google Colab	*Cloud-based development environment used for high-performance model training and experimentation.*
Random Forest Classifier	*The specific Machine Learning algorithm selected for fetal risk classification due to its high accuracy and low computational cost.*
Flask (Python Framework)	*Lightweight web framework used to create the RESTful API that connects the AI model with the mobile application.*
Pickle (.pkl)	*Serialization format used to save the trained model for efficient loading and offline inference.*
VS Code	*Integrated Development Environment (IDE) used for backend coding and API debugging.*
Postman	*Tool used for testing API endpoints and verifying JSON request/response structures.*

Functional Requirements

1. **Data Input:** The system must accept 21 CTG parameters (including Baseline Heart Rate, Accelerations, Uterine Contractions) as input.
2. **Risk Prediction:** The system must accurately classify the fetal state into three categories: **Normal, Suspect, or Pathological**.
3. **Explainability:** The system must identify and display the specific reasons for a high-risk prediction (e.g., "Prolonged Decelerations Detected").
4. **Confidence Scoring:** The system should provide a probability score (percentage) indicating the confidence level of the prediction.
5. **Offline Capability:** The prediction engine must function locally on the device or via a local server without requiring active internet connectivity.

Non-Functional Requirements

1. **Accuracy:** The AI model must achieve a testing accuracy of **90% or higher** on the validation dataset.
2. **Performance:** The risk prediction result must be generated within **2 seconds** after data submission.
3. **Reliability:** The system must provide consistent results for the same input values (Deterministic behavior).

4. **Resource Efficiency:** The trained model file size should be optimized (lightweight) to ensure it runs smoothly on mobile devices with limited RAM.
5. **Scalability:** The backend API should be able to handle multiple concurrent requests without crashing.

System Requirements

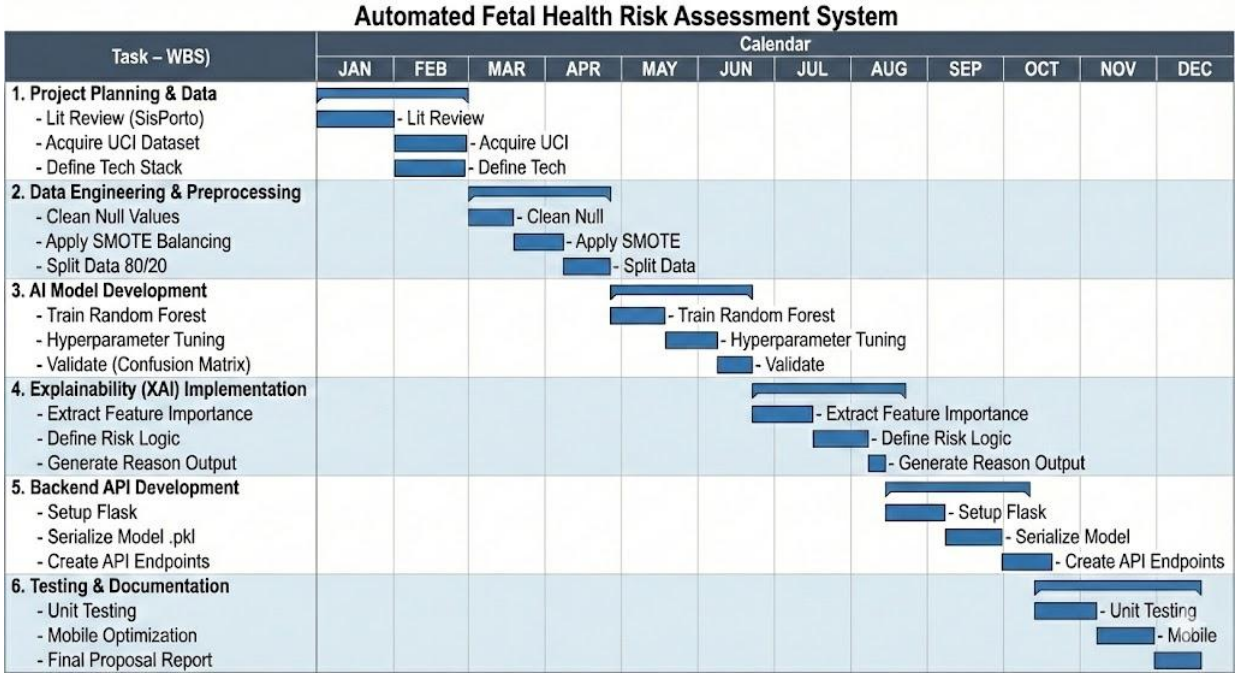
Hardware Requirement

1. Development: Laptop with at least 8GB RAM (for running ML training locally if needed) or access to Cloud GPUs (Google Colab).
2. Testing: Android Smartphone (Mid-range) to test the mobile integration and response speed.

Software Requirements

1. **Language:** Python 3.8 or higher.
2. **Libraries:** Scikit-learn, Pandas, NumPy, Flask.
3. **Tools:** Visual Studio Code, Postman, Git.

Figure 4 Gantt Chart



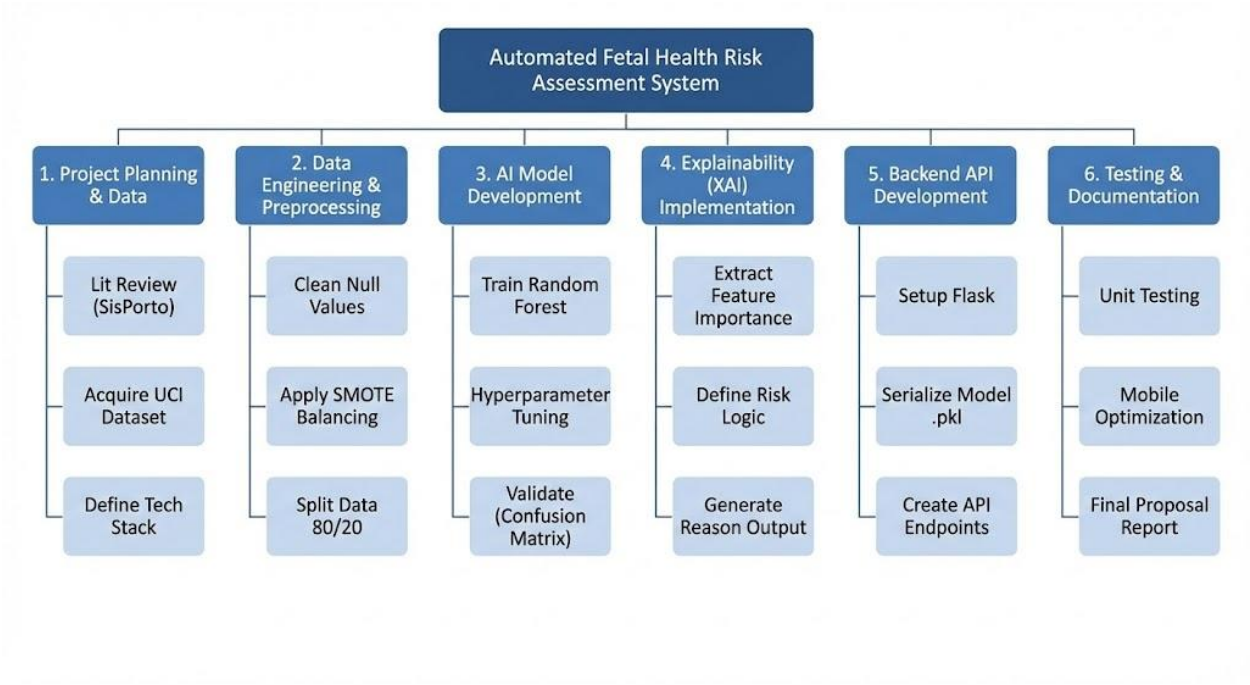


Figure 5 Work Breakdown Structure (WBS)

Description of Personal and Facilities

Supervisors

Our supervisor Mr. Didula Thanaweera Araghchi and Co-Supervisor Mrs. Narmada Gamage, who are experts in the technical field, will provide guidance in software development and technology integration.

External Supervisor

Our external supervisor is a midwife who has been in the field for over 26 years. She shared real experiences, how she manages mothers before, during and after birth. Her knowledge helped us understand the real problems midwives face daily. With her guidance, we were able to plan the app feature more practically and usefully.

Facilities

The project is carried out with support from the Humanities and Nursing Departments of SLIIT. Access is provided to medical books, maternal health guidelines, and computer labs for app development and testing. This setup ensures that the mobile app is developed in line with real-world midwifery practices in Sri Lanka.

Sample of Midwifery Students

A group of 15–20 midwifery and nursing students will take part in the testing phase. They will use the app, give feedback on its features, and help improve its design and content. This hands-on testing also gives students practical learning opportunities.

Device Access

The app will be tested on commonly used Android smartphones, including low and mid-range devices. This ensures it works well for public health midwives in both urban and rural areas. The testing will also cover local language support, offline access, and data entry functions.

Budget and Budget Justification

Table 3 Budget Table

Expenses	
Requirement	Cost (Rs.)
Mobile Devices for Testing (Android + iOS)	180,000
Backend Server Hosting (1 year)	60,000
Firebase / Cloud Database (annual)	45,000
Software Licenses (Figma, IDEs, API tools)	30,000
Data Analysis Tools (Python Libraries, Power BI Pro)	50,000

Reference List

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